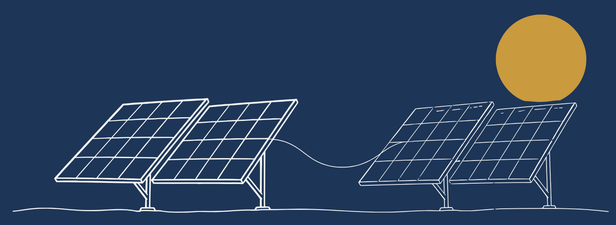
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LEARNING WHEN TO TRANSFER: ROBUST SOLAR POWER FORECASTING UNDER CAPACITY MISMATCH

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PROBLEM STATEMENT & MOTIVATION

New PV plants start with no generation history. A model trained on a larger plant learns its output magnitude and overfits to it when transferred to a smaller one.

14 MW
Source plant

→

1.4 MW
Target plant

Magnitude overfitting under a ten-fold capacity gap.

OBJECTIVES

1. Investigate **zero-target-label**, unsupervised transfer.
2. **Decouple scale** through per-unit normalization.
3. **Remove the daily cycle** with the Clear-Sky Index (CSI).
4. **Align dynamics** with a conditional domain adversarial network (CDAN).

KEY CONTRIBUTIONS & DATA

1. **Failure-mode characterization**
   Shows that magnitude overfitting causes the collapse when source and target capacities differ.
2. **Physics-guided domain alignment**
   Per-unit scaling and the CSI remove capacity and daily-cycle effects before CDAN aligns what remains.
3. **Transfer boundary delineation**
   Tests transfer within India and across continents to Australia, where the seasons are reversed.

India
Mature source
Northern Hemisphere

→

Australia
Cross-continental target
Southern Hemisphere

PROPOSED METHODOLOGY & ARCHITECTURE

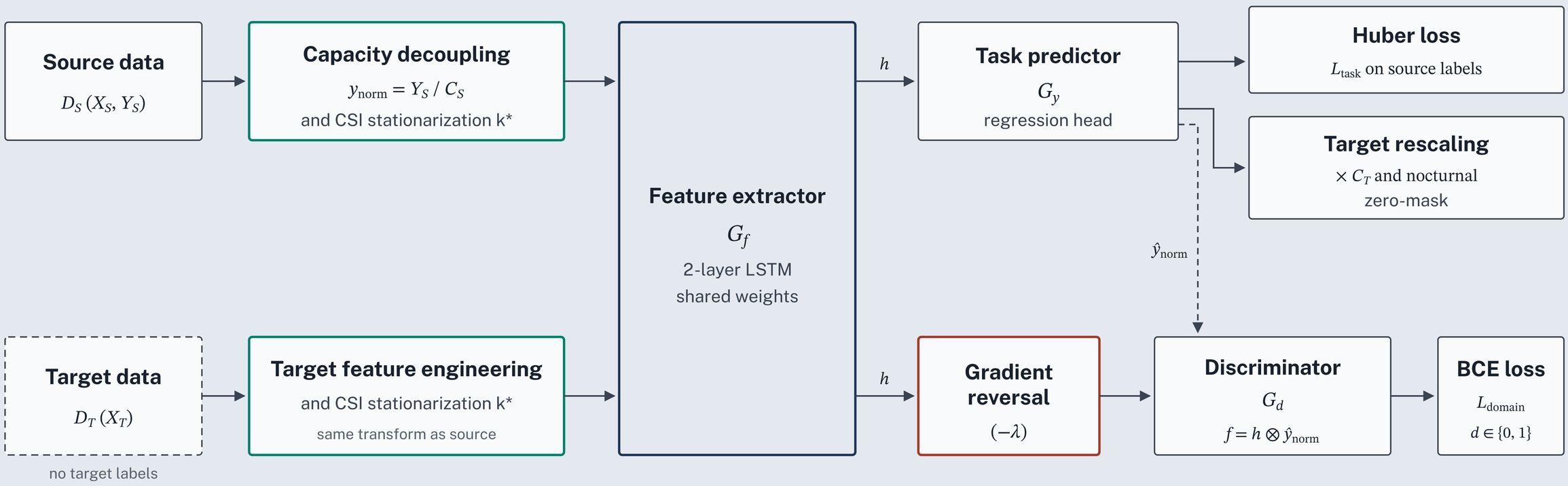


Fig. 1. Capacity decoupling and CSI stationarization feed a shared 2-layer LSTM. A gradient reversal layer trains the conditional domain discriminator, so source and target features align without target labels.

EXPERIMENTAL BENCHMARK & TRACKING

Model (R^2)	Task 1 India	Task 2 Australia	Task 3 Australia
A1 · Source-only LSTM	**0.8325**	–	–
A3 · CDAN + LSTM	**0.8329**	–	–
A4 · Transformer	–	**0.5428**	**0.5428**
Proposed · CDAN + LSTM + CSI	**0.8325**	**0.5428**	–

Place the 48-hour tracking figure from the paper here

Fig. 2. 48-hour tracking: proposed (CDAN + CSI) against ground truth. The uncalibrated baseline exceeds 10,000 kW (off-scale); hard-zero gating holds output at zero at night.

DISCUSSION & FUTURE WORK

Key insight

Adversarial alignment works well within a region. Across continents (India to Australia) it causes negative transfer ($R^2 < 0$), because the seasons are out of phase. An unadapted Transformer is more robust there and keeps positive skill.

Intra-regional CDAN
Aligned representation

Cross-hemisphere CDAN
Negative transfer

Conclusion

- Per-unit scaling and nocturnal zero-masking fix the magnitude collapse of the uncalibrated baseline.
- Defines where adaptation helps: within a region, but not across hemispheres.
- Establishes a zero-target-label transfer benchmark under capacity mismatch.
- Future work: fine-tuning after commissioning, once the first target labels arrive.

Generative AI and LLM tools assisted in preparing this poster.

REFERENCES & CONTACT

[1] D. Belay et al., “Learning When to Transfer: Robust Solar Power Forecasting Under Capacity Mismatch,” NEECON 2026.
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